BT404
Food Biotechhnology
Current Finall Term Solved Spring 2022
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QUESTION 1

1.Marketing during the time period of baby boomer 5

• Baby boomers (born between 1946-1964),
• Most powerful market
• Developing products for this market has gained prominence

2. Write the structure trailer 3

3.Bioreactor monitoring systems and design 5

• The stirred tank bioreactor design is the most common fermentor and consists of agitator, baffles, aeration sparger for aerobic culture growth, sterilizable monitoring probes for pH, dissolved oxygen, temperature, and antifoam, filling and draining ports, and often culture medium sterilization capabilities in the reactor tank.
• Reactor agitation is essential for temperature control, pH adjustments, oxygen absorption into the liquid medium, overall culture health, and mixing of any required additions of substrate, and nutrients.
Fermentation health requires real time monitoring system

4.Functionality of Starches 5

• Crystalline packing of double helices of starch chains from amylose fiber shows diffraction pattern through wide angle x-ray scattering.
• When starch granules are heated sufficiently in excess water, this pattern is lost and an endothermic transition is observed.
• Double helical order is additionally lost during this endothermic event, and it is this loss and not the loss of crystalline order that is believed to be responsible for the endothermic event
• This loss of granular, crystalline, and double helical order is called gelatinization, and it is irreversible with respect to the three dimensional granule organization.
• Subsequent to gelatinization, starch chains of both amylose and amylopectin may organize into new double helical and crystalline structures in a process called retrogradation.
• In its native granular form mixed with water, starch makes a high solids mixture, useful in making batters and doughs.
• During heating, starch granules swells 30-fold its dry weight.

• This process is reversible provided temperatures and pressures are lower than those that lead to disruption of the organized intramolecular glucan chain associations within the granule
• Above these temperatures and pressures, the granules will undergo irreversible changes and glucan chains will dissociate, permitting further swelling.
• additional heating and shear forces, swollen hydrated granules will eventually collapse to form an unorganized paste of starch molecules.
• results in a thickening effect, imparting texture and structure.
• In functional terms, this process of starch granule swelling and dissociation is • The glucan polymers of a gelatinized starch can associate with each other or other components of food to impart stickiness, tackiness or a rubbery texture.
• amylose also readily leaches out of the granule during gelatinization and interacts with the food components.
• In mild cooking conditions, amylopectin does not so readily leach and hence is enriched inside the gelatinized granule.
• extreme cooking conditions, the amylopectin is much more dispersed, resulting in products with a tacky or sticky character known as gelatinization
• Once a cooked starch is allowed to cool, the glucan chains begin to reassociate in a process resulting in a change in the functional characteristics, including a decrease in paste clarity and gelation of the paste. In functional terms, this process of reorganization is known as retrogradation.
• Normal starches are generally recognized for their ability to retrograde within hours and usually form opaque gels.
• Retrogradation of amylopectin gels occurs during days or weeks of storage.
• As a consequence of their different molecular weights and chain length profiles, the rates of retrogradation of amylose and amylopectin are not the same.
• The rapid setting of the structure of breads is believed to be due to rapid (within seconds or minutes) amylose retrogradation to form a network structure.
• Starch functionality is therefore a consequence of the degree of gelatinization and is influenced by retrogradation, time, temperature, concentration, etc.

5.Purpose of food additive. 3

• Food additives have been defined as chemical substances deliberately added to foods, directly or indirectly, in known and regulated quantities, for purposes of assisting in the processing of foods; preservation of foods; or in improving the flavor, texture, or appearance of foods
• An additive may be reactive or inactive; it may be nutritive or nonnutritive; it should be neither toxic or hazardous.
• Some substances, such as pesticides, are added to foods unintentionally, and these are, of course, undesirable, and may be hazardous to health

Name the pathways of Starch-Sugar Balance 2

• Sucrose is formed from starch by an i**rreversible pathway**
• Starch is formed from sucrose by separate, but likewise i**rreversible, routes**

6. Enhancement of bowelmotility 2

Enhancement of bowel motility, (relief from constipation & reduction of inflammatory or allergic reactions) by:

• Restoration of the homeostasis of the immune system.

• Regulation of cytokine synthesis.

7. Millard reaction 2

• Non enzymatic browning of the product occurs at high temperature, when reducing sugars combine with the alpha-amino groups of amino acids Resulting in dark and more bitter products that are unacceptable to the consumer

• Known as Maillard reaction

types of meat fermentation products

salami, ham, and sausages

why sterilizatiin is necessary for food

Heat sterilisation : foods are heated at a sufficiently high temperature and for a sufficiently long time to destroy microbial and enzyme activity.

Sterilised foods have a shelf life in excess of six months at ambient temperatures.

symptoms caused by enterotoxin

Strains of Staphylococcus aureus produce an enterotoxin.

It can cause:

1. Salivation
2. Nausea
3. Vomiting
4. Abdominal cramps
5. Diarrhoea.

The time between the consumption of the food and the appearance of symptoms can range from 1–6 hours.

Recovery takes about a day or two and mortality is very low.

process of converting milk into yogurt (fermentation)

Curd (Yogurt):

• Curds is a very popular milk product in India.

• It ranks second to milk in the extent of consumption.

• Milk is boiled and cooled to about 50°C and a teaspoon of curd from an earlier batch of curd is added and mixed thoroughly.

• The lactic acid bacteria present in this sample of curd curdle the milk.

• They utilize the lactose in milk and break it down to lactic acid

. • The formation of lactic acid increases the acidity of milk. When the pH reaches 4.6, the milk protein, casein, coagulates as curd
. • Since the process is gradual, the milk serum is bound in the mass of coagulated proteins.
• The optimum temperature for the formation of curd is 35- 40°C.
• The time needed for curd formation varies from 8-12 hours depending on atmospheric temperature.
• Once made, curd keeps well at refrigerator temperature of 2–3 days. It is used as dressing on salads made from fresh vegetables.
• It combines well with plain cooked rice.
• Preparation of curd is one method of prolonging the shelf-life of milk by several days

factors on which heat exchanger depend

The main factors affecting the performance heat exchanger tubes include: fluid velocity, tube diameter, converter tube shape (U or cross shape), converter layout order (horizontal or vertical), and Venting valve, material of tube sheet and channel and location order of input channel.

transgenic plants

Transgenic plants are plants into which one or more genes from another species have been introduced into the genome, using genetic engineering processes. Techniques include the biolistic method—in which a heavy metal is coated with plasmid DNA is shot into cells—and Agrobacterium tumefaciens mediated transformation.

Infrared drying

Infrared drying works when the infrared radiation penetrates the material, and it is converted into heat energy and thus removes water more evenly. Infrared drying allows a more uniform distribution of heat than convection drying, and therefore, a higher-quality product is obtained

macro environment marketing

• Macro environment is formed of some societal forces that influence its marketing activities Demographic, Economic, Natural, Technological, Political, Cultural

1- write two methods for counting non pathogen from milk

•Non Pathogens in milk Spoilage of refrigerated milk consists usually of bitter, rancid, fruity flavours
Different methods to count non pathogens in milk such as:
Viable count(pour plate method).
Dye reduction test Direct (clump) microscopic count

2- characteristics of lipids

Oils and fats belong to naturally occurring substances called lipids.
The common characteristics of lipids are:

1. They are soluble in organic solvents (ether, acetone etc.)
2. They are insoluble in water
3. Most of them are derivatives of fatty acids

3- write two methods for straining (is trha ka hi tha kuch)

An acid-fast stain is able to differentiate two types of gram-positive cells: those that have waxy mycolic acids in their cell walls, and those that do not. Two different methods for acid-fast staining are the **Ziehl-Neelsen technique and the Kinyoun technique**. Both use carbolfuchsin as the primary stain

4- what are four P's of marketing (long)

The basic principles of marketing include the 4 P's
• Product
• Price
• Place
• Promotion

5- the processes by which cearls are processed

• A number of processed cereal products are made from grains
• Include products like:
• Cracked wheat
• Semolina
• Atta
• Maida
• Rice flakes
• Puffed rice
• Processes involved:
• Grinding
• Roasting
• Pounding
• Processes increase surface area of product exposed to atmosphere
• Processes:
• Decrease preparation time
• Reduce shelf life
• Whole grain shelf life is almost a year or more
• Processed cereal foods shelf life varies from two weeks to a few months

6- name vegetables that harbor lactic acid bacteria (long)

Almost all vegetables can be fermented through natural processes, because they harbor many types of lactic acid bacteria e.g. sauerkraut , reddish, turnips , beets , beans, peas , corn etc.

7- what are antioxidants

Antioxidants are food additives used to stabilize foods against oxidation.
• Oxidative quality changes in foods include:
•the development of rancidity from the oxidation of unsaturated fats resulting in off-odors and offflavors and
•discoloration from oxidation of pigments or other components of the food.
• Antioxidants function in different ways, the purpose of each is to prevent, delay, or minimize the oxidation of the food to which they are added.
• antioxidants make their combination with oxygen.
• or prevent oxygen from reacting with components of the food, as in a hermetically sealed container

8- what are processes of fermentation (long)

As the desirable microorganisms grow in specific conditions on selected foods, they utilize some nutrients and produce some end products.
1) Alcoholic Fermentation :- (Alcohol end product)
2) Acetic Acid Fermentation (Acetic acid end product)
3) Lactic Acid Fermentation (Lactic Acid end product)

Alcoholic Fermentation:
In this fermentation Alcohol is produced as a end product by the activity of micro-organisms (yeast)
Mostly used by Bakery industry for special aroma

Acetic Acid Fermentation :-
In this fermentation Acetic acid is produced as end product by the activity of Acetobacter bacteria on sugary / alcoholic food stuff
Used for the production of vinegar

Lactic Acid Fermentation
In this fermentation glucose and other six Carbon sugars are converted into cellular energy and metabolite lactate via Lactic acid producing bacteria as starter culture
Mostly used for the production of yoghurt from milk

9- Why is fermentation used more in industrial Biotechnology than last 20 years?

It is the result of the corn syrup sweetener industry and computer technology.
• Corn is 70% starch, and when dried to 19%, it can be stored for more than two years.
• glucose syrups from corn is consistent, low cost supply of substrate.
• For example, corn syrups containing 19% dextrose, 14% maltose, 12% maltotriose, and 55% higher saccharides are used to control microbial growth rates and biological heat production in many fermentations.
• Thus, glucose is the platform chemical used for the microbial production of organic acids, amino acids, vitamins, and more
. • In fact some food grade fermentation facilities have located adjacent to a corn sweetener facility to permit glucose syrup to be piped directly to their fermentors.
• Glucose corn syrup is also shipped via truck or rail cars as liquid or dried product.

• Furthermore, the advent of computer process controls of industrial scale fermenters has removed many of the fears associated with commercial scale fermentation by providing reliable and easy to operate dissolved oxygen, pH, foam, temperature, and sterilization controls of the process.

10- under what conditions food and vegetables are kept

Fruits and vegetables must be stored under

Low oxygen

High CO_2

Proper storage temperature

Internet Source

A general rule for vegetables is that cool-season crops should be stored at cooler temperatures (32 to 35°F), and warm-season crops should be stored at warmer temperatures (45 to 55°F

The most basic rule must be always followed: store raw products below, never above, your cooked or ready-to-eat products. Critical Control Point. Keep foods **4°C (39°F) or colder**, the safe temperature for refrigerated storage